

Worksheet -2



Name: BINA THAPA

Module Title: Programming in C++

Student id- 23085129

Task 1: Basic student grading system prototype using classes and objects. [20 Marks]

Write a program that manages a simple student grade calculator with the following requirements. Create a Student class that has:

1. Student name (string)
2. Three subject marks (integers)
3. A basic member function to calculate average

The program should:

1. Accept student details (name and marks) from user input
2. Calculate and display:
 1. Total marks
 2. Average marks
 3. Grade (A for $\geq 90\%$, B for $\geq 80\%$, C for $\geq 70\%$, D for $\geq 60\%$, F for $< 60\%$)
3. Display a message if any mark is below 0 or above 100

➤ **Answer:**

```
#include <iostream>
using namespace std;
class Student{
protected:
    string Firstname,Lastname;
    int subject1,subject2,subject3;
    char Grade;
    float Total,Average;
public:
    Student()
    {
        Firstname="";
        Lastname="";
        subject1=0;
        subject2=0;
        subject3=0;
        Total=0.0;
        Average=0.0;
    }
    void getinput()
```

```

{
    cout<<"Enter the Student Firstname: "<<endl;
    cin>>Firstname;

    cout<<"Enter the Student Lastname: "<<endl;
    cin>>Lastname;

    cout<<"Enter the marks for subject1: "<<endl;
    cin>>subject1;
    cout<<"Enter the marks for subject2: "<<endl;
    cin>>subject2;
    cout<<"Enter the marks for subject3: "<<endl;
    cin>>subject3;

}

void check()
{
    if(subject1<0 || subject1>100 || subject2<0 ||subject2>100 ||subject3<0
||subject3>100 )
    {

        cout<<"The marks cannot exceed 100 or fall behind 0"<<endl<<"Check
your input"<<endl;
        getinput();
    }
}

void cal()
{
    Total= subject1+subject2+subject3;
    Average=Total/3;
    Gradecal();
}

void Gradecal()
{
    if (Average>=90)
    {
        Grade='A';
    }
    else if (Average>=80)
    {
        Grade='B';
    }
}

```

```

        else if(Average>=70)
        {
            Grade='C';
        }
        else if(Average>=60)
        {
            Grade='D';
        }
        else
        {
            Grade='F';
        }
    }
}

```

```

void Display()
{
    cout<<"-----REPORT CARD -----"<<endl;
    cout<<"Student Name: "<<Firstname<<" "<<Lastname<<endl;
    cout<<"Total Marks: "<<Total<<endl;
    cout<<"Average Marks: "<<Average<<"%"<<endl;
    cout<<"Grade: "<<Grade<<endl;
    cout<<"-----"<<endl;
}

```

```
};
```

```

int main()
{
    Student s1;
    s1.getinput();
    s1.check();
    s1.cal();
    s1.Display();

    return 0;
}

```

❖ Output

```
"C:\Users\beena\OneDrive\De  X + v
Enter the Student Firstname:
Maya
Enter the Student Lastname:
Ale
Enter the marks for subject1:
90
Enter the marks for subject2:
80
Enter the marks for subject3:
85
-----REPORT CARD -----
Student Name: Maya Ale
Total Marks: 255
Average Marks: 85%
Grade: B
-----
Process returned 0 (0x0)   execution time : 21.997 s
Press any key to continue.
|
```

Task 2: Programming assignments: All questions are mandatory

1. Write a program with a class Circle having:
 1. Private member: radius (float)
 2. A constructor to initialize radius
 3. A friend function compareTwoCircles that takes two Circle objects and prints which circle has the larger area

➤ **Answer:**

```
#include <iostream>
using namespace std;
class Circle2;
class Circle1
{
private:
    float a;
public:
    void getdata(float r1)
    {
```

```

        a=r1;
    }

    friend void compareTwoCircles(Circle1 &r1,Circle2 &r2);
};
class Circle2
{
private:
    float b;
public:
    void getdata(float r2)
    {
        b=r2;
    }

    friend void compareTwoCircles(Circle1 &r1,Circle2 &r2);
};

void compareTwoCircles(Circle1 &r1,Circle2 &r2)
{
    float Area1=3.14*r1.a*r1.a;
    float Area2=3.14*r2.b*r2.b;

    cout<<" Area of first circle: "<<Area1<<endl;
    cout<<" Area of second circle: "<<Area2<<endl;

    if(Area1>Area2)
    {
        cout<<"First circle has larger area!"<<endl;
    }
    else if(Area1<Area2)
    {
        cout<<"Second circle has larger area!"<<endl;
    }
    else
    {
        cout<<"Both circle have equal area!"<<endl;
    }
}

void TestCircle()
{

```

```
float r1, r2;
cout << " Circle Comparison "<<endl;
cout << "Enter radius for Circle1: ";
cin >> r1;
cout << "Enter radius for Circle2: ";
cin >> r2;

Circle1 c1;
Circle2 c2;
c1.getdata(r1);
c2.getdata(r2);
compareTwoCircles(c1, c2);
}

int main()
{
    TestCircle();

    return 0;
}
```

❖ Output


```

        return a;
    }
    else
    {
        return b;
    }
}
int findMax(int a,int b, int c)
{
    if(a>=b && a>=c)
    {
        return a;
    }
    else if(b>=a && b>=c )
    {
        return b;
    }
    else
    {
        return c;
    }
}

float findMax(float a, int b)
{
    if (a>b)
    {
        return a;
    }
    else
    {
        return b;
    }
}

void TestMax()
{
    int I1,I2,I3;
    float F1,F2;
    int I;
    float F;

    cout << " Enter two integers: "<<endl;

```

```

    cin >> I1 >> I2;
    cout << "Max of " << I1 << " and " << I2 << " is: " << findMax(I1,I2) << endl;

    cout << "Enter two float numbers: "<<endl;
    cin >> F1 >> F2;
    cout << "Max of " << F1 << " and " << F2 << " is: " << findMax(F1, F2) <<
endl;

    cout << "Enter three integers: "<<endl;
    cin >> I1 >> I2 >> I3;
    cout << "Max of " << I1 << ", " << I2 << ", and " << I3 << " is: " <<
findMax(I1, I2, I3) << endl;

    cout << "Enter a float and an integer: "<<endl;
    cin >> F >> I;
    cout << "Max of " << F << " and " << I << " is: " << findMax(F,I) << endl;

}
};
int main()
{
    calculate c1;
    c1.TestMax();
    return 0;
}

```

❖ Output

```
"C:\Users\beena\OneDrive\De... X + v
Enter two integers:
2
3
Max of 2 and 3 is: 3
Enter two float numbers:
4.5
5.3
Max of 4.5 and 5.3 is: 5.3
Enter three integers:
8
4
2
Max of 8, 4, and 2 is: 8
Enter a float and an integer:
3.4
5
Max of 3.4 and 5 is: 5

Process returned 0 (0x0)   execution time : 14.667 s
Press any key to continue.
}
}
}
```

Task 3: Basics of File Handling

1. Write a program that reads the titles of 10 books (use an array of 150 characters) and writes them in a binary file selected by the user. The program should read a title and display a message to indicate if it is contained in the file or not.

➤ **Answer:**

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;
```

```
class BookManager
{
    char titles[10][150];
    char filename[100];
```

```
public:
    void getFilename()
    {
```

```

    cout << "Enter the binary file name: ";
    cin >> filename;
}

void inputTitles()
{
    cout << "Enter 10 book titles:\n";
    for (int i = 0; i < 10; i++) {
        cout << "Title " << (i + 1) << ": ";
        cin >> titles[i];
    }
}

void writeToFile()
{
    ofstream out(filename, ios::binary);
    for (int i = 0; i < 10; i++)
    {
        out.write(titles[i], sizeof(titles[i]));
    }
    out.close();
    cout << "Book titles saved to file."<<endl;
}

void searchTitle()
{
    char searchbook[150];
    cout << "Enter the book title to search: "<<endl;
    cin >> searchbook;

    ifstream in(filename, ios::binary);
    char temp[150];
    bool found = false;

    while (in.read(temp, sizeof(temp))) {
        if (strcmp(temp, searchbook) == 0) {
            found = true;
            break;
        }
    }

    in.close();
}

```

```

        if (found)
            cout << "Book found in the file.";
        else
            cout << "Book not found in the file.";
    }
};

int main() {
    BookManager b;
    b.getFilename();

    int choice;

    while (true)
    {
        cout << " Menu: "<<endl;
        cout << "1. Write 10 book titles to binary file "<<endl;
        cout << "2. Search for a book title "<<endl;
        cout << "3. Exit"<<endl;
        cout << "Enter your choice: "<<endl;
        cin >> choice;

        if (choice == 1)
        {
            b.inputTitles();
            b.writeToFile();
        } else if (choice == 2)
        {
            b.searchTitle();
        } else if (choice == 3)
        {
            cout << "Exiting program..."<<endl;
            break;
        } else
        {
            cout << " Try again!"<<endl;
        }
    }

    return 0;
}

```

❖ **Output**



"C:\Users\beena\OneDrive\De



Enter the binary file name: book

Menu:

1. Write 10 book titles to binary file
2. Search for a book title
3. Exit

Enter your choice:

1

Enter 10 book titles:

Title 1: snow

Title 2: pines

Title 3: red

Title 4: bill

Title 5: atlas

Title 6: summit

Title 7: sapiens

Title 8: hope

Title 9: suffering

Title 10: shadow

Book titles saved to file.

Menu:

1. Write 10 book titles to binary file
2. Search for a book title
3. Exit

Enter your choice:

|

```

Menu:
1. Write 10 book titles to binary file
2. Search for a book title
3. Exit
Enter your choice:
2
Enter the book title to search:
snow
Book found in the file. Menu:
1. Write 10 book titles to binary file
2. Search for a book title
3. Exit
Enter your choice:
2
Enter the book title to search:
bina
Book not found in the file. Menu:
1. Write 10 book titles to binary file
2. Search for a book title
3. Exit
Enter your choice:

```

2. Create a program that:

1. Reads student records (roll, name, marks) from a text file
2. Throws an exception if marks are not between 0 and 100
3. Allows adding new records with proper validation
4. Saves modified records back to file

➤ Answer:

```

#include <iostream>
#include <fstream>
using namespace std;

```

```

class Students
{
    int Rollno;
    string Name;
    float Marks;

public:
    Students()
    {
        Rollno = 0;
        Name = "";
        Marks = 0;
    }

    Students(int r, string n, float m)
    {
        if (m < 0 || m > 100)
        {
            throw string("Invalid input!");
        }
        Rollno = r;
        Name = n;
        Marks = m;
    }

    void getinput()
    {
        cout << "Enter Student Details: "<<endl;
        cout << "Student Roll No: "<<endl;
        cin >> Rollno;
        cout << "Student Name: "<<endl;
        cin >> Name;
        cout << "Marks obtained: "<<endl;
        cin >> Marks;

        if (Marks < 0 || Marks > 100) {
            throw string(" Invalid input!");
        }
    }

    void display()
    {
        cout << "Roll No: " << Rollno<< " "<< endl;
    }
}

```

```

        cout<< "Name: " << Name <<" "<<endl ;
        cout << "Marks: " << Marks << " "<<endl;
        cout<<"===== "<<endl;
    }

    void readfile()
    {
        ifstream file("Marks.txt");

        while (file >> Rollno >> Name >> Marks) {
            if (Marks < 0 || Marks > 100) {
                throw string("Invalid marks found in file!");
            }
            display();
        }

        file.close();
    }

    void writefile()
    {
        ofstream out("Marks.txt", ios::app);
        out << Rollno << " " << Name << " " << Marks << endl;
        out.close();
    }
};

int main()
{
    try
    {
        Students s1;
        s1.getinput();
        s1.writefile();
        s1.readfile();
        cout<<"New record saved!!"<<endl;
    }
    catch(string x)
    {
        cout<<x<<endl;
    }
    return 0;
}

```

❖ Output

```
"C:\Users\beena\OneDrive\De  X + v
Enter Student Details:
Student Roll No:
56
Student Name:
Jhon
Marks obtained:
90
Roll No: 77
Name: Ani
Marks: 99
=====
Roll No: 5
Name: Maya
Marks: 89
=====
Roll No: 56
Name: Jhon
Marks: 90
=====
New record saved!!

Process returned 0 (0x0)   execution time : 7.996 s
Press any key to continue.
|
```